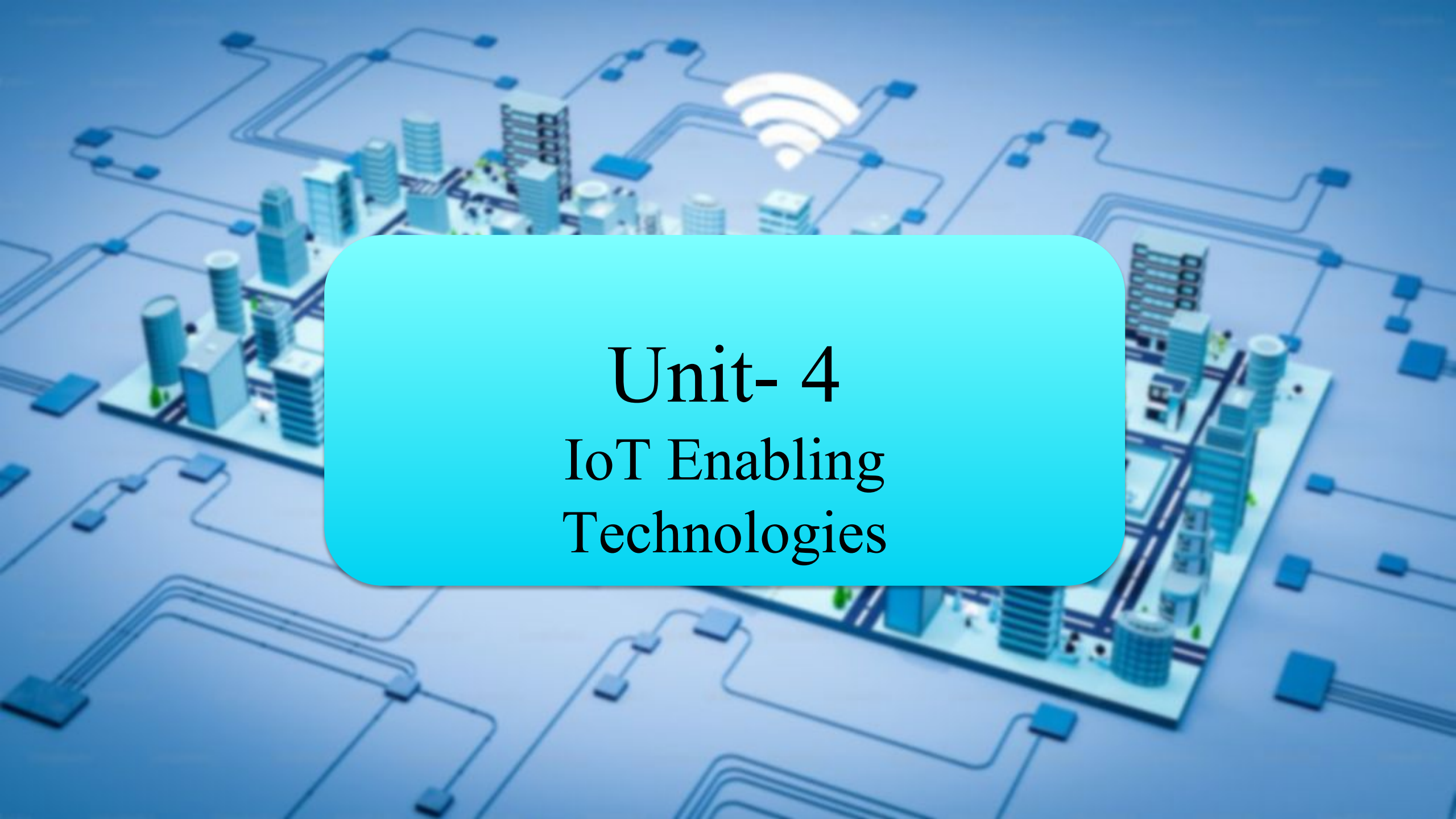


The background is a stylized illustration of a smart city. It features a grid of roads with various buildings, including skyscrapers and smaller structures. A network of blue lines, representing data or communication paths, crisscrosses the scene, connecting different parts of the city. A prominent white Wi-Fi symbol is located in the upper center, indicating wireless connectivity. The overall color palette is dominated by light blue and white, giving it a clean, technological feel.

Unit- 4

IoT Enabling Technologies

Introduction to Wireless Sensor Network (WSN)

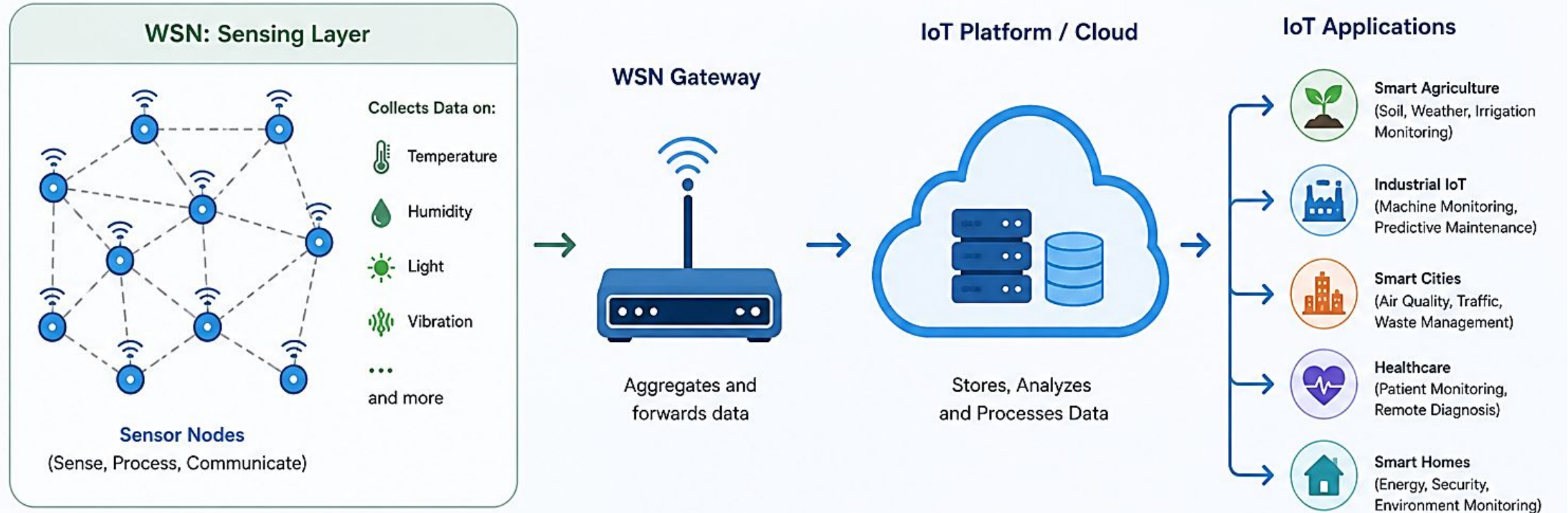
- A Wireless Sensor Network (WSN) is a collection of small, low-cost, low-power sensor nodes that are deployed in an environment to monitor physical or environmental conditions such as temperature, humidity, pressure, light, motion, sound, and pollution.
- These sensor nodes communicate wirelessly and cooperate to send collected data to a central station called the Base Station (Sink Node). WSN is one of the most important enabling technologies of the Internet of Things (IoT).

Features of WSN

- Self-organizing network
- Wireless communication
- Low power consumption
- Distributed sensing
- Real-time monitoring
- Scalability



Role of WSN in IoT



Key Roles of WSN in IoT

Data Collection

WSN sensors collect real-time data from the physical environment.



Reliable Communication

WSN provides multi-hop wireless communication for efficient and reliable data transmission.



Energy Efficiency

WSN nodes are designed for low power consumption, enabling long-term operation.



Data Aggregation

WSN aggregates and filters data to reduce redundancy and network traffic.



Scalability

WSN supports large-scale deployments, making IoT solutions scalable and cost-effective.



Context Awareness

WSN provides real-time, context-aware data for smarter decision-making in IoT applications.

Components of Wireless Sensor Network

1. Sensor Nodes

- These are the basic units of a WSN that collect data from the environment.

2. Sensor

- Detects physical parameters such as:
- Temperature
- Humidity
- Light
- Motion
- Gas concentration

3. Microcontroller (Processing Unit)

- Processes sensed data
- Controls node operations
- Executes communication protocols

4. Transceiver (Communication Unit)

- Sends and receives data wirelessly
- Uses technologies such as ZigBee, Wi-Fi, Bluetooth, LoRa

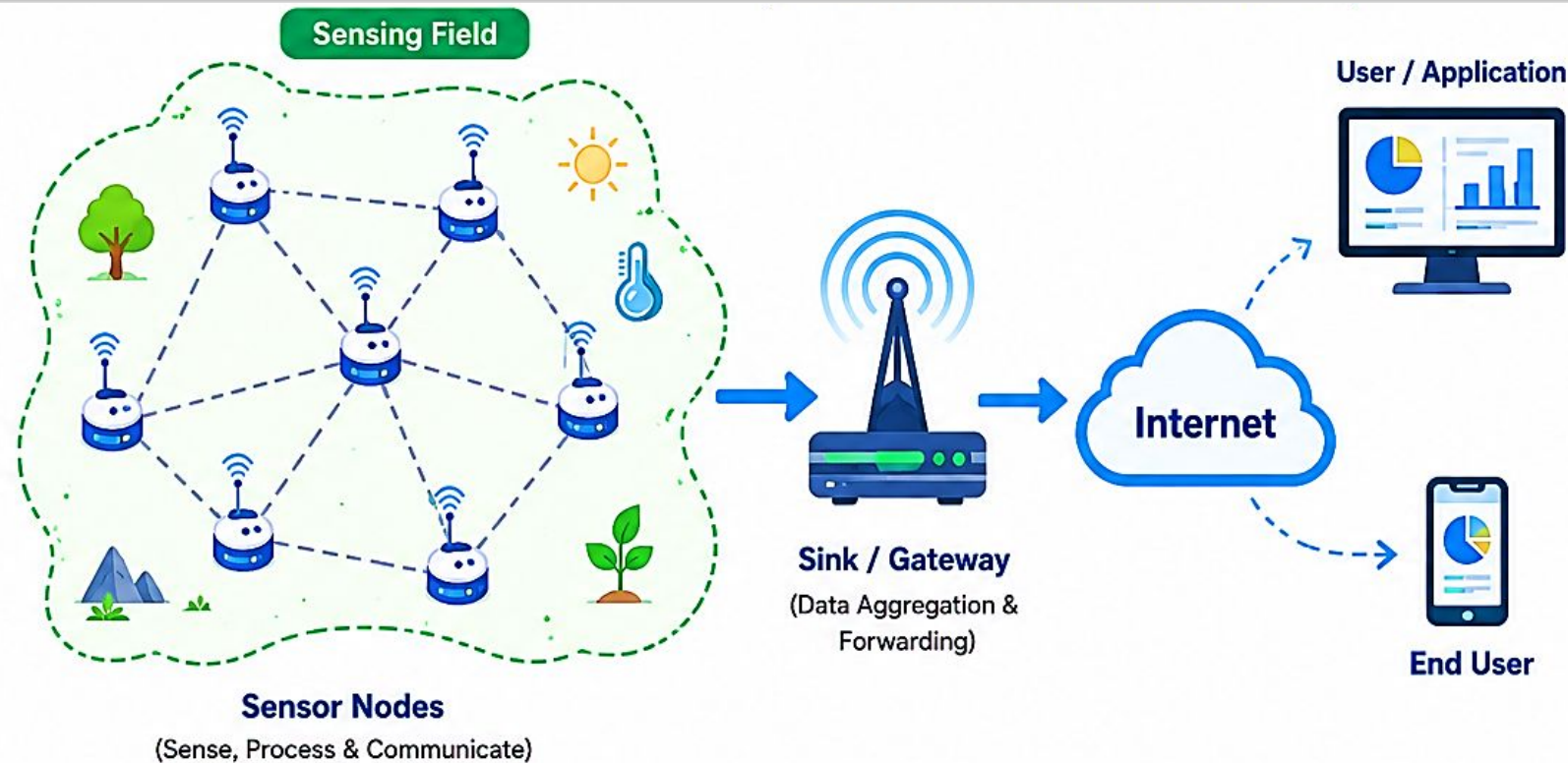
5. Power Unit

- Battery or energy harvesting source
- Supplies power to the node

6. Base Station (Sink Node)

- Collects data from sensor nodes
- Connects WSN to the Internet or cloud

Components of Wireless Sensor Network



Key Characteristics of WSN Components



Low Power Consumption



Wireless Communication



Distributed Operation



Reliable & Fault Tolerant



Scalable & Flexible



Real-time Monitoring



Together, these components enable WSN to collect, transmit and utilize data efficiently.

1



Sensor Nodes

Sense data from the environment and transmit it to other nodes.

2



Sink / Gateway

Collects data from sensor nodes and forwards it to the base station / cloud.

3



Communication Network

Provides multi-hop wireless communication between nodes and gateway.

4



Base Station / Server

Stores, processes and manages data from the network.

5

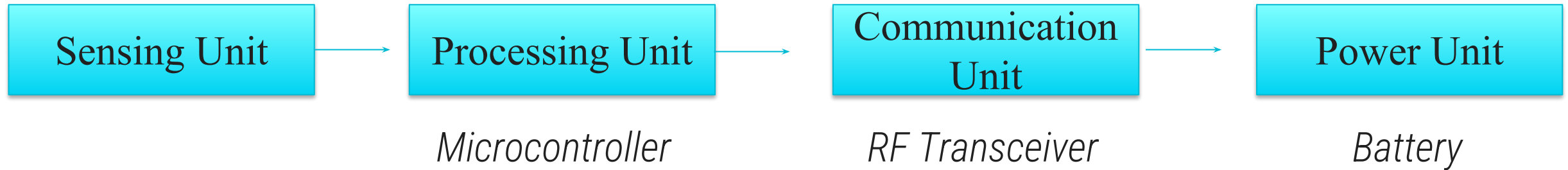


User / Application

Accesses data and uses it for monitoring, analysis and decision making.

Sensor Node Architecture

- A typical sensor node consists of:



Sensing Unit

- Contains sensors and ADC (Analog to Digital Converter)
- Collects environmental data

Processing Unit

- Processes and stores data
- Makes decisions before transmission

Communication Unit

- Transfers data to neighboring nodes or sink node

Power Unit

- Provides energy for sensing and communication

Node Behaviour in WSN

- Sensor nodes perform various operations to ensure efficient network functioning.

1. Sensing

- Continuously monitors the environment.
- Collects data using sensors.

2. Processing

- Filters and processes collected data.
- Removes redundant information.

3. Communication

- Transmits data to neighboring nodes or sink node.

4. Routing

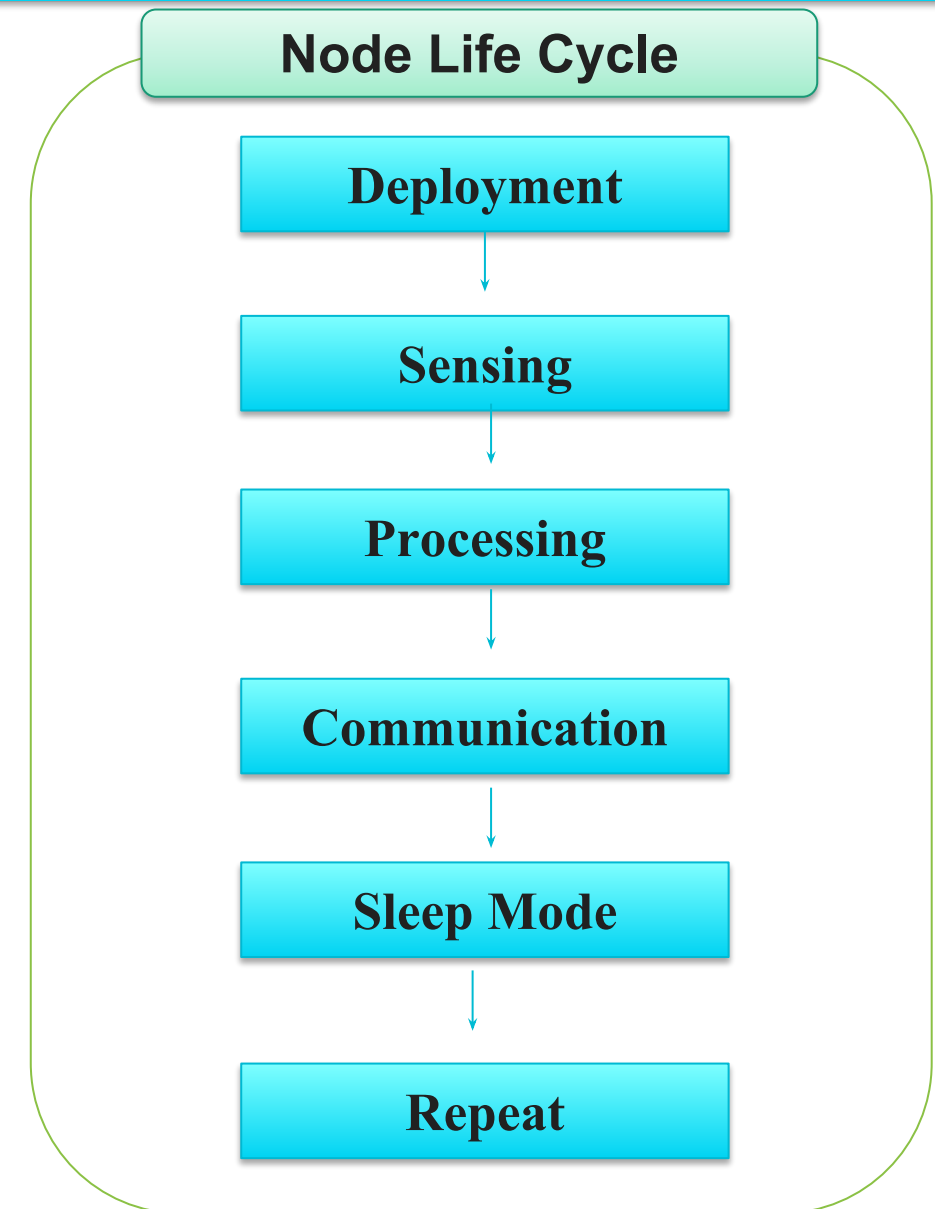
- Finds optimal path for data transmission.

5. Sleep Mode

- Conserves battery power.
- Activates only when required.

6. Self-Organization

- Automatically joins and configures the network.



Challenges of Wireless Sensor Networks

1. Energy Consumption

- Nodes operate on batteries.
- Frequent communication drains power.

2. Limited Processing Power

- Small microcontrollers have limited resources.

3. Limited Memory

- Storage capacity is very low.

4. Communication Range

- Wireless signals have limited coverage.

5. Security Issues

- Vulnerable to attacks such as:
- Data interception
- Node capture
- Denial of Service (DoS)

6. Scalability

- Managing thousands of nodes becomes difficult.

7. Reliability

- Nodes may fail due to battery depletion or environmental conditions.

8. Environmental Factors

- Temperature, rain, dust, and obstacles affect performance.

9. Data Redundancy

- Multiple nodes may send duplicate data.

10. Network Congestion

- Large amounts of traffic can reduce performance.

Applications of WSN in IoT

1 Smart Agriculture

- Soil moisture monitoring
- Irrigation control
- Crop health monitoring



2 Smart Home



- Temperature control
- Security systems
- Smart lighting

3 Healthcare Monitoring



- Patient health tracking
- Wearable devices
- Remote monitoring

4 Environmental Monitoring



- Air quality monitoring
- Forest fire detection
- Weather monitoring

5 Industrial Automation



- Machine condition monitoring
- Predictive maintenance

6 Smart City



- Traffic monitoring
- Smart parking
- Waste management

Applications of WSN in IoT

7 Military Applications



- Battlefield surveillance
- Intrusion detection

8 Disaster Management



- Earthquake monitoring
- Flood detection
- Early warning systems

9 Transportation



- Vehicle tracking
- Fleet management

10 Energy Management



- Smart grid monitoring
- Power consumption analysis



Introduction to Cloud Computing

What is Cloud Computing?

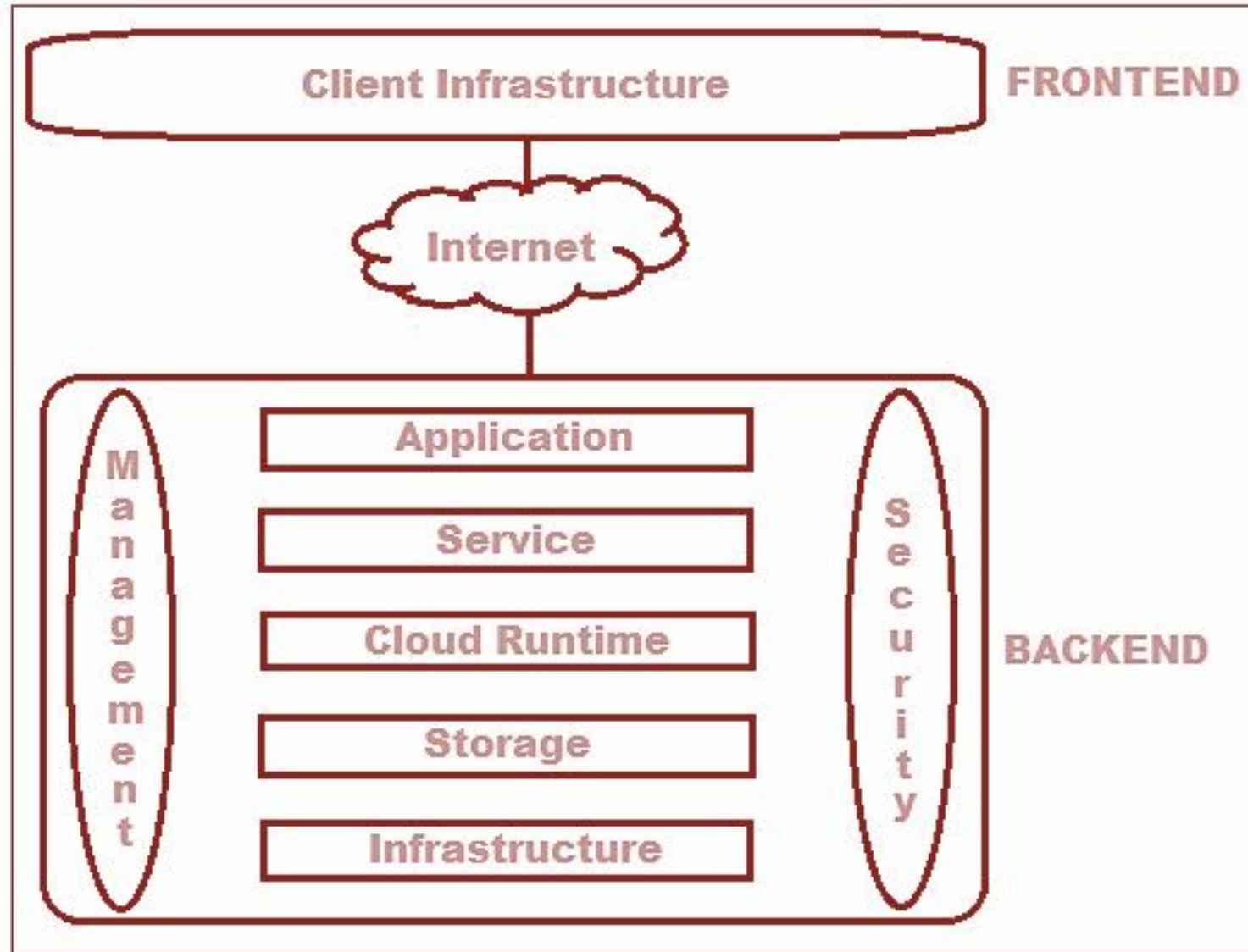
- Cloud Computing is a technology that enables users to access computing resources such as servers, storage, databases, networking, software, and analytics over the Internet without owning or maintaining physical infrastructure.
- Traditionally, organizations had to purchase servers, install software, maintain data centers, and employ IT staff. Cloud computing eliminates these requirements by providing resources on demand through cloud service providers.
- *"Cloud Computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort."*

Why Cloud Computing?

Organizations adopt cloud computing because it offers:

- Reduced infrastructure costs
- High scalability
- Better performance
- Faster deployment
- Global accessibility
- Business continuity
- Automatic updates

Architecture of Cloud Computing



Components of Cloud Computing

1. Front-End

The front-end is the user side of cloud computing.

Components:

- Web browsers
- Mobile applications
- User interfaces
- Thin clients

Examples:

- Chrome Browser
- Mobile Banking Apps

Components of Cloud Computing

2. Back-End

- The back-end consists of the cloud infrastructure.

Components:

- Servers
- Storage systems
- Databases
- Virtual machines
- Security systems

3. Network

- The Internet connects users to cloud services.

Functions:

- Data transmission
- Communication
- Service access

Characteristics / Features of Cloud Computing

1. On-Demand Self-Service

- Users can access computing resources automatically whenever required without contacting service providers.

Example

- A company can create a virtual server within minutes.

Benefits

- Faster provisioning
- Reduced waiting time

Characteristics / Features of Cloud Computing

2. Broad Network Access

- Cloud services are accessible through the Internet using different devices.

Supported Devices

- Desktop Computers
- Laptops
- Smartphones
- Tablets
- IoT Devices

Benefits

- Anywhere access
- Device independence

Characteristics / Features of Cloud Computing

3. Resource Pooling

- Cloud providers pool resources and share them among multiple customers.

Resources Shared

- CPU
- Memory
- Storage
- Network bandwidth

This concept is called Multi-Tenancy.

Benefits

- Efficient utilization
- Reduced costs

Characteristics / Features of Cloud Computing

4. Rapid Elasticity

- Cloud resources can be scaled up or down automatically.

Example

- An e-commerce website may require:
- 10 servers during normal days
- 100 servers during festivals

Cloud automatically adjusts resources.

Benefits

- Improved performance
- Cost savings

Characteristics / Features of Cloud Computing

5. Measured Service

- Cloud systems monitor resource usage.
- Users pay only for:
 - Storage used
 - Computing hours consumed
 - Network bandwidth used

Example

- Using 100 GB storage costs less than 500 GB storage.

Characteristics / Features of Cloud Computing

6. High Availability

- Cloud services are available 24×7.

Techniques Used

- Redundant servers
- Backup systems
- Failover mechanisms

Benefits

- Minimal downtime

Characteristics / Features of Cloud Computing

7. Scalability

- Ability to increase or decrease resources according to workload.

Types

- Vertical Scaling
 - Increase power of existing server.
 - Example:
 - RAM from 8 GB to 32 GB
- Horizontal Scaling
 - Add more servers.
 - Example:
 - Increase from 5 servers to 20 servers

Characteristics / Features of Cloud Computing

8. Reliability

- Cloud providers maintain multiple copies of data.

Benefits

- Data protection
- Business continuity

9. Security

- Security mechanisms include:
 - Encryption
 - Authentication
 - Authorization
 - Firewalls
 - Intrusion Detection Systems

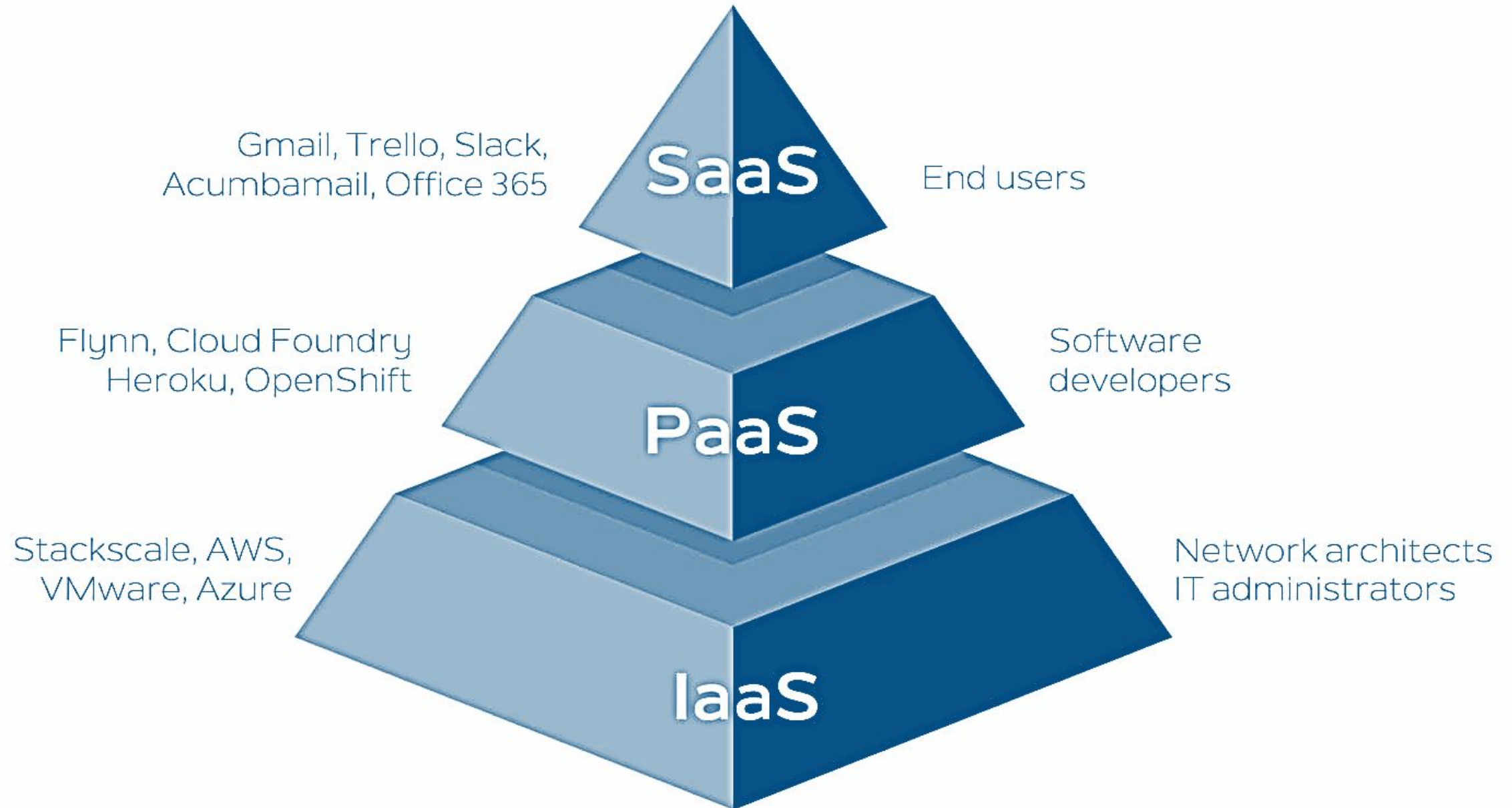
10. Cost Efficiency

- Organizations avoid purchasing:
 - Hardware
 - Software licenses
 - Data center infrastructure

Result:

- Lower capital expenditure (CAPEX)

Cloud Service Models



Cloud Service Models

1) Infrastructure as a Service (IaaS)

- Infrastructure as a Service (IaaS) is a cloud computing model that provides virtualized computing resources such as servers, storage, networking, and operating systems over the Internet.
- Instead of purchasing and maintaining physical hardware, users can rent these resources from a cloud provider on a pay-as-you-go basis.

Services Provided by

IaaS Virtual Machines (VMs)

- Virtual computers that run operating systems and applications.
- Users can choose CPU, RAM, and storage as needed.
- Example: Creating a Windows or Linux server in the cloud.

Cloud Service Models

2. Storage

- Provides data storage facilities.
- Can be used for backups, databases, files, and applications.
- Highly scalable and secure.

3. Networking

- Virtual networks, load balancers, firewalls, and IP addresses.
- Helps connect cloud resources securely.

4. Servers

- Cloud providers offer virtual servers instead of physical servers.
- Users can start, stop, and configure servers anytime.

Cloud Service Models

Example

Netflix

- Netflix uses cloud infrastructure from Amazon Web Services (AWS).
- Instead of building thousands of physical servers worldwide, Netflix rents:
 - Virtual Machines for running streaming applications.
 - Storage for movies and TV shows.
 - Networking services to deliver content globally.
 - Cloud servers that automatically scale when millions of users watch videos.
- **Benefit:** Netflix can serve millions of users without owning and managing all the physical hardware.

2. Platform as a Service

(PaaS) Platform as a Service (PaaS) is a cloud computing model that provides a complete platform for developing, testing, deploying, and managing applications over the Internet.

- The cloud provider manages the underlying infrastructure, operating system, databases, and development tools, allowing developers to focus only on building applications.

Services Provided by PaaS

1. Application Development Environment

- Provides tools and frameworks for coding applications.
- Supports multiple programming languages such as Java, Python, PHP, and .NET.

Cloud Service Models

2. Database Management

- Built-in databases for storing and managing application data.
- No need to install or maintain database servers.

3. Operating System Management

- The cloud provider manages the operating system.
- Developers do not need to handle updates or patches.

4. Deployment Tools

- Easy deployment of web and mobile applications.
- Applications can be published with minimal configuration.

5. Testing and Debugging Tools

- Tools for testing applications before deployment.
- Helps identify and fix errors efficiently.

Example

Google App Engine

- Google provides Google App Engine, a PaaS platform.
- A developer can:
 - Write application code.
 - Upload it to Google App Engine.
 - Google automatically handles servers, operating systems, storage, and scaling.

Benefit: Developers focus only on coding while Google manages the platform.

3. Software as a Service

(SaaS)

Software as a Service (SaaS) is a cloud computing model in which software applications are delivered over the Internet.

- Users can access the software through a web browser without installing or maintaining it on their own computers.
- The cloud provider manages everything, including infrastructure, platform, updates, security, and maintenance.

Services Provided by SaaS

1. Ready-to-Use Software

- Applications are available immediately through the Internet.
- No installation is required on the user's device.

Cloud Service Models

2. Automatic Updates

- Software updates and bug fixes are handled by the provider.
- Users always access the latest version.

3. Data Storage

- User data is stored securely in the cloud.
- Accessible from anywhere with an Internet connection.

4. Collaboration Features

- Multiple users can work together in real time.
- Ideal for teams and organizations.

5. Accessibility

- Can be accessed from computers, tablets, and smartphones.
- Supports remote work and learning.

Cloud Service Models

Example

Gmail

- Google Gmail is a SaaS application.
- Users can:
 - Send and receive emails.
 - Store emails in the cloud.
 - Access their accounts from any device using a web browser.
- Google manages:
 - Servers
 - Storage
 - Security
 - Software updates
- **Benefit:** Users simply use the email service without worrying about installation or maintenance.

Cloud Deployment Models

- Cloud Deployment Models define how cloud infrastructure is deployed, managed, and accessed by users. They determine who owns the cloud resources, where they are located, and who can use them.
- There are four main cloud deployment models:
 - 1.Public Cloud
 - 2.Private Cloud
 - 3.Hybrid Cloud
 - 4.Community Cloud

Big Data Analytics & IoT Domains

What is Big

Data? Big Data Analytics is the technique of analyzing large volumes of structured, semi-structured, and unstructured data to extract meaningful insights for business and scientific applications.

Why Big Data is

Important?

Organizations use Big Data Analytics to:

- Improve business decisions
- Predict future trends
- Enhance customer experience
- Detect fraud and security threats
- Optimize operations
- Support scientific research

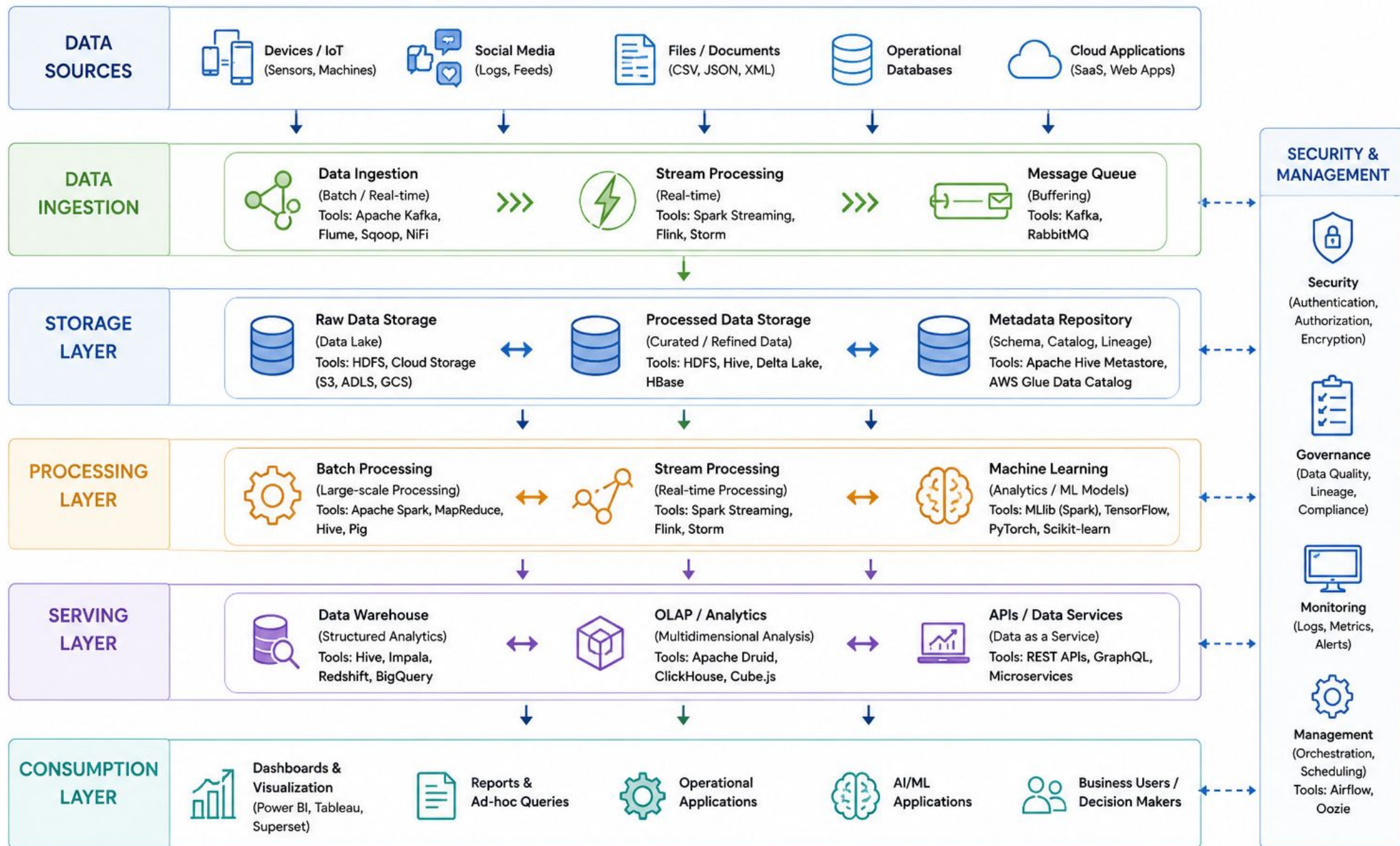
Example

- Netflix analyzes viewing history to suggest movies and web series.
- Google analyzes search patterns to improve search results.

Characteristics of Big Data



Big Data Architecture



Data Handling in Big Data

- Data Handling refers to collecting, storing, managing, and processing large volumes of data efficiently.

1. Data Collection

- Data is gathered from multiple sources.

Sources

- IoT Sensors
- Social Media
- Websites
- Mobile Apps
- Enterprise Systems

Example:

- A smart city collects data from traffic sensors, weather stations, and CCTV cameras.

Data Handling in Big Data

2. Data Storage

- Large-scale storage systems are required.

Technologies

- Hadoop Distributed File System (HDFS)
- Cloud Storage
- Data Warehouses

Benefits:

- Scalability
- Reliability
- Fault Tolerance

Data Handling in Big Data

3. Data Cleaning

- Raw data often contains errors.

Cleaning Activities

- Removing duplicates
- Correcting errors
- Filling missing values

Example:

- Removing duplicate customer records from a database.

4. Data Integration

- Combining data from multiple sources into a unified format.

Example:

- A hospital combines patient data from laboratory systems, pharmacy systems, and electronic health records.

Data Handling in Big Data

5. Data Processing

- Processing converts raw data into useful information.

Types

- Batch Processing
- Processes large data sets at scheduled intervals.

Example:

- Monthly payroll processing.

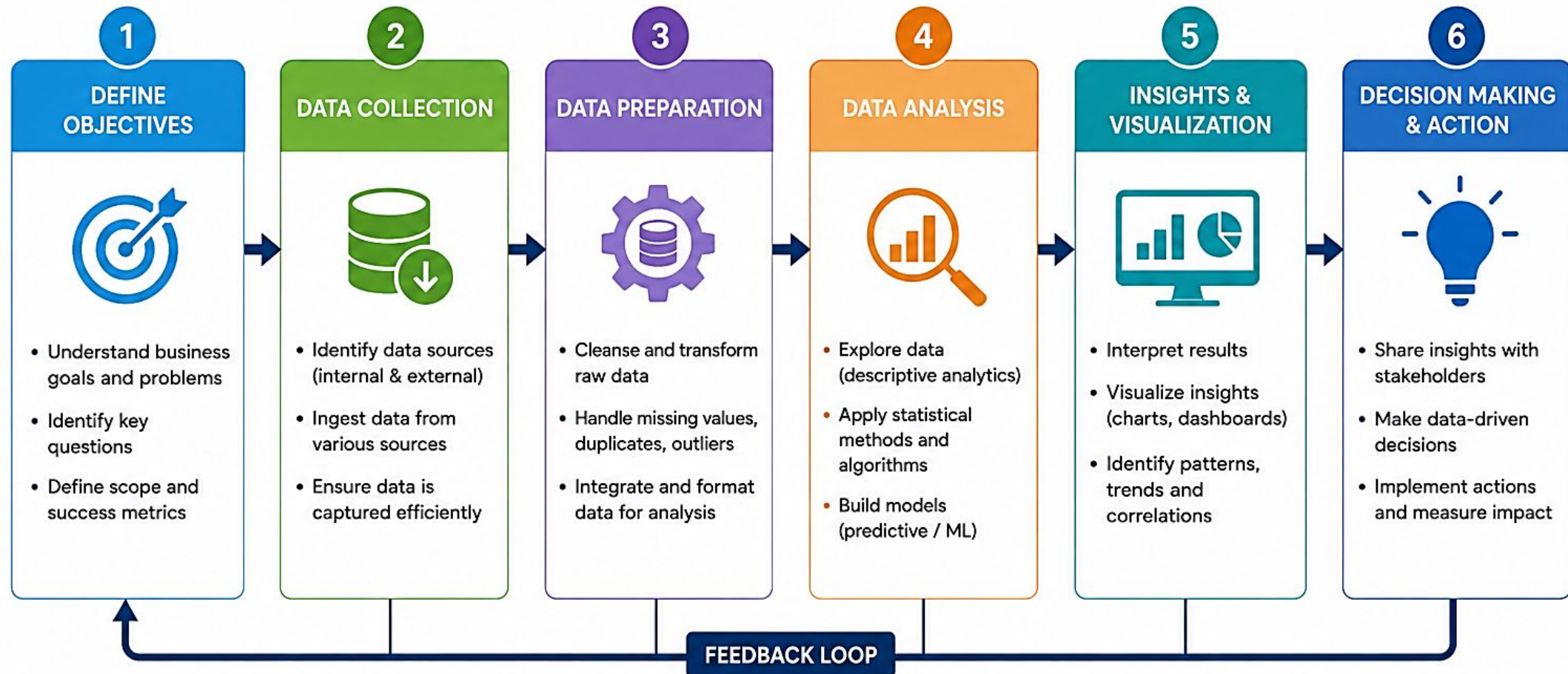
Real-Time Processing

- Processes data immediately after generation.

Example:

- Credit card fraud detection.

Big Data Analytics Process



Evaluate Results → Refine Approach → Improve Continuously



Applications of Big Data Analytics / IoT Domains



1. RETAIL & E-COMMERCE

- Customer segmentation
- Personalized recommendations
- Demand forecasting
- Inventory optimization
- Market basket analysis



2. BANKING & FINANCIAL SERVICES

- Fraud detection
- Risk assessment
- Customer segmentation
- Algorithmic trading
- Credit scoring



3. HEALTHCARE

- Predict patient outcomes
- Disease prediction
- Personalized medicine
- Healthcare analytics
- Fraud and abuse detection



4. MANUFACTURING

- Predictive maintenance
- Quality control
- Supply chain optimization
- Production planning
- Equipment performance analysis



5. TRANSPORTATION & LOGISTICS

- Route optimization
- Fleet management
- Real-time tracking
- Demand forecasting
- Delivery time prediction



6. TELECOMMUNICATIONS

- Customer churn analysis
- Network optimization
- Fraud detection
- Usage pattern analysis
- Personalized offers



7. CYBERSECURITY

- Threat detection
- Anomaly detection
- User behavior analytics
- Fraud prevention
- Security intelligence



8. MARKETING & ADVERTISING

- Customer segmentation
- Campaign optimization
- Sentiment analysis
- Market trend analysis
- ROI measurement



9. GOVERNMENT & PUBLIC SECTOR

- Smart city initiatives
- Public safety analytics
- Policy impact analysis
- Tax fraud detection
- Resource optimization



10. EDUCATION

- Student performance analysis
- Personalized learning
- Dropout prediction
- Curriculum improvement
- Administrative analytics



11. MEDIA & ENTERTAINMENT

- Content recommendation
- Audience analysis
- Ad targeting
- Trend analysis
- Viewership prediction



12. ENERGY & UTILITIES

- Smart grid analytics
- Energy usage forecasting
- Equipment maintenance
- Outage prediction
- Sustainability analytics

Case Study: Smart Home Automation

- Smart Home Automation is an IoT-based system that enables users to remotely monitor and control household appliances using smartphones, computers, or voice assistants.
- The system improves comfort, security, and energy efficiency.

Objectives

- Automate household devices
- Reduce energy consumption
- Improve security
- Enable remote access

Case Study: Smart Home Automation

Components Used

Hardware

- **Sensors**

- Temperature Sensor
- Motion Sensor (PIR)
- Smoke Sensor
- Light Sensor

Actuators

- Relay Modules
- Smart Switches
- Smart Locks

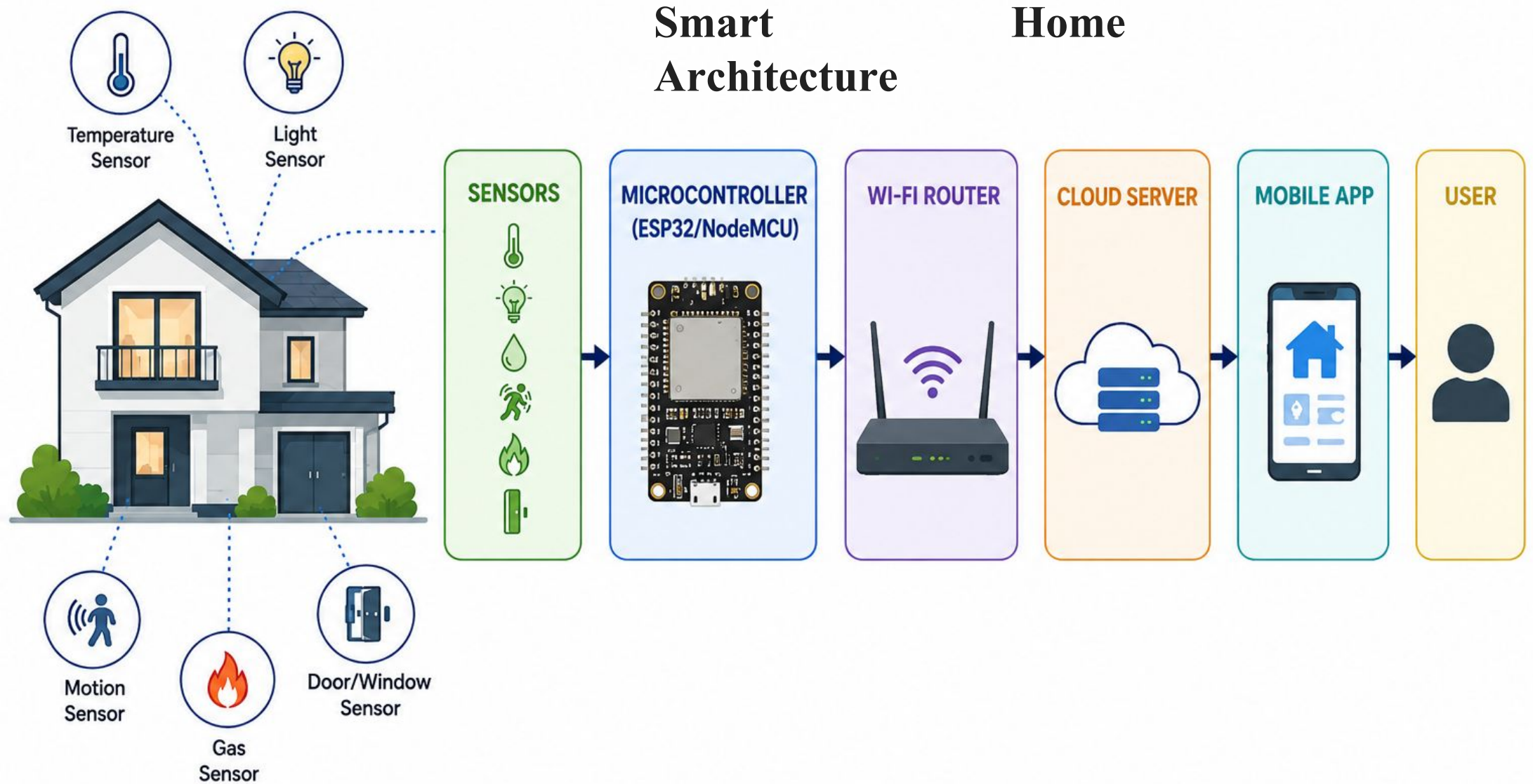
Controller

- Arduino UNO
- NodeMCU ESP8266
- ESP32

Communication

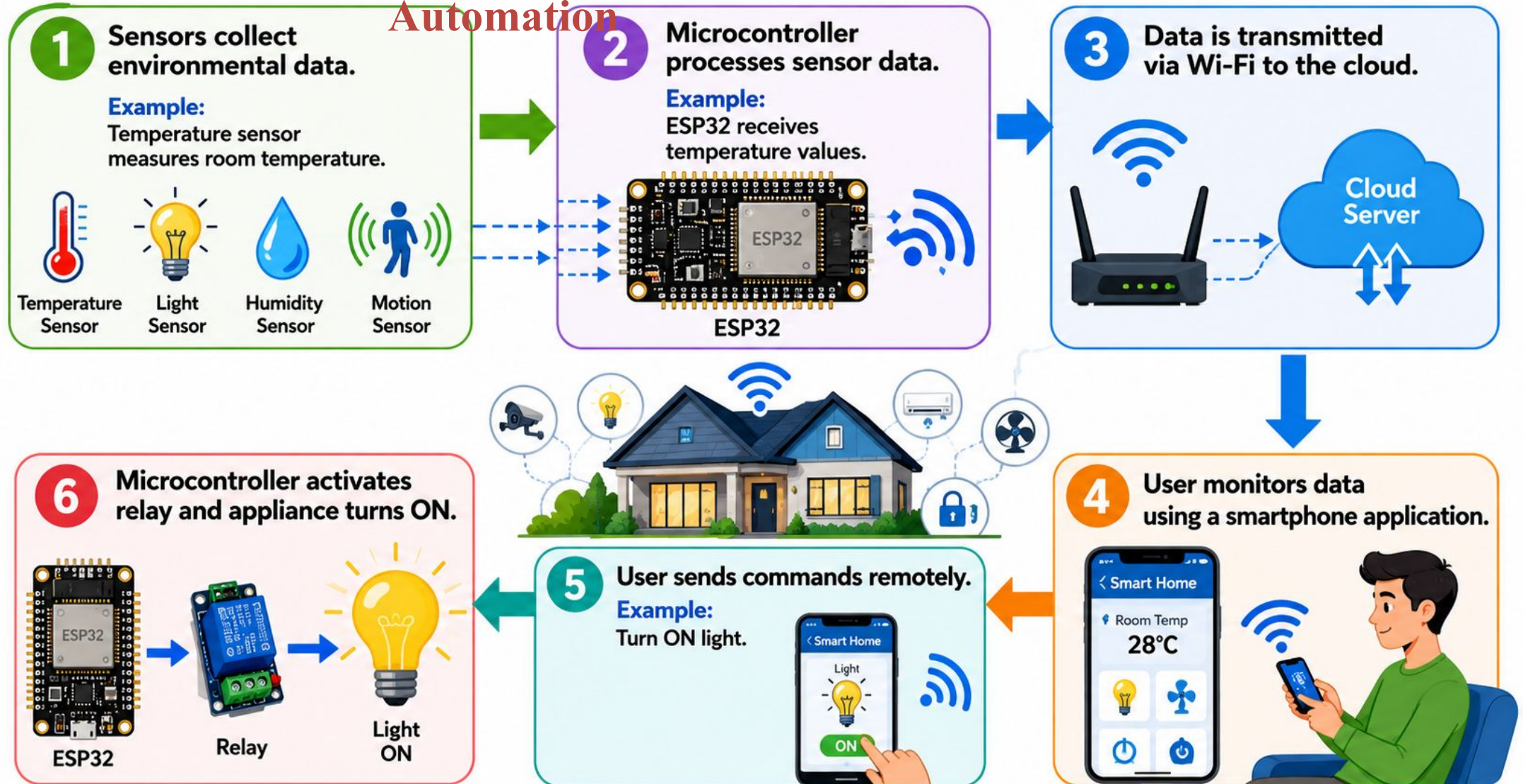
- Wi-Fi
- Bluetooth
- ZigBee

Case Study: Smart Home Automation



Case Study: Smart Home Automation

Working of Smart Home Automation



Case Study: Smart Home Automation

Smart

Home



SMART LIGHTING

Automatically turns lights ON/OFF.



BENEFITS

-  Energy savings
-  Convenience



SMART SECURITY

Includes:

- Smart locks
- CCTV monitoring
- Motion detection



BENEFITS

-  Enhanced security
-  Intrusion alerts



SMART CLIMATE CONTROL

Automatically controls:

- Fans
- Air conditioners
- Heaters



BENEFITS

-  Comfortable environment
-  Reduced electricity usage



SMART APPLIANCES

Examples:

- Smart TV
- Smart Refrigerator
- Smart Washing Machine



Case Study: Smart Home Automation

Advantages of Smart Home Automation

- Remote monitoring
- Energy efficiency
- Improved security
- Convenience
- Better resource management
- Reduced electricity bills

Case Study: Smart Home Automation

Challenges in Smart Home Automation

Security Risks

- Hackers may gain access to devices.

Privacy Concerns

- Personal data may be exposed.

Network Dependency

- Requires reliable internet connection.

Compatibility Issues

- Different manufacturers may use different standards.

High Initial Cost

- Installation and devices can be expensive.